SVETHAA L

AI/ML Engineer | Python, AI/ML Fundamentals & Data-Driven Systems

svethaalingaswaran05@gmail.com | LinkedIn | GitHub | Coimbatore, Tamil Nadu

SUMMARY

Final-year B. Tech Information Technology student (2023–2027) at Kumaraguru College of Technology with a strong foundation in AI/ML fundamentals and hands-on experience building data-driven backend systems using Python, Flask, MySQL, and MongoDB. Approached an AI Agent solution for customer support at a national-level Capgemini hackathon and applied data-backed assessment frameworks during a digital-transformation internship. Eager to apply strong problem-solving skills to real-world machine learning and computer vision challenges.

TECHNICAL SKILLS

Languages: Java, Python, C, SQL

AI/ML & Data: AI/ML Fundamentals, Big Data Analytics (Fundamentals), OpenCV, Data Handling, Streamlit, Tableau

Backend & Frameworks: Python Flask, RESTful API Design

Databases: MySQL, MongoDB

Cloud: AWS Cloud Foundations, IoT Basics

Tools: Git, Docker(basics), RapidMiner AI studio, Mendix, Google Colab.

PROJECTS

Production Line Instability Prediction — Altair AI Studio (RapidMiner) · Random Forest

- Built a machine learning-based production line monitoring system to classify manufacturing conditions as stable or unstable, supporting early identification of potential process instability.
- Performed data preparation, attribute selection, feature and target variable creation.
- Applied 5-fold stratified cross-validation and grid-based hyperparameter optimization to improve model performance, achieving an overall accuracy of 98.68%.

AeroX – Aircraft Software Change Impact Assessment — Mendix · Microflows · Domain Modeling

- Built a Mendix-based application to assess software-change impacts across aircraft, configurations, systems, and test cases.
- Designed structured entity relationships and automated impact tracing using Mendix microflows to generate affected impact items.
- Implemented review/approval workflows and analytics dashboards for tracking software-change status and impact levels.

Forest Fire Detection-Comparative analysis of SVM and CNN — Python · OpenCV · TensorFlow

- Built an image-based fire detection system comparing classical ML (SVM + HOG features) against deep learning (CNN) for binary classification of forest images.
- Owned the image preprocessing and CNN pipeline — used OpenCV for decoding/resizing, TensorFlow/Keras for model serving, and tuned classification threshold via Precision-Recall analysis (0.5 to 0.182) to cut false negatives.
- Deployed both models via Flask with real-time confidence scoring and comparative accuracy analysis between the two approaches.

INTERNSHIP EXPERIENCE

Digital Transformation Intern – Dx-EDGE — March 2026 – Present

- Conduct digital maturity assessments for MSME clients using the Dx-Edge framework, evaluating readiness across technology, operations, and digital adoption.
- Perform gap analysis on organizational digital capabilities and contribute to assessment frameworks presented to stakeholders.

Siemens | Software Industry Readiness Program — July 2026 – Present

- Pursuing industry-focused training in Data Science, including coursework in Machine Learning and Data Engineering, with emphasis on developing practical technical and analytical skills.
- Developing multiple practical projects, including a major capstone project, integrating the knowledge and skills acquired throughout the Software Industry Readiness Program.
- Participated in the SynTech Data Science Contest and other hands-on learning activities, applied data science concepts to real-world problem statements and projects.

EDUCATION

B. Tech Information Technology – Kumaraguru College of Technology, Coimbatore 2023 – 2027
CGPA: 8.3

CERTIFICATIONS

- Software Engineering ALM – Infosys springboot
- Machine Learning and Data Engineering Professional – Altair Engineering
- Rapid Developer - Mendix
- AWS Cloud Foundations — AWS Academy

ACHIEVEMENTS & ACTIVITIES

- Active Member of SIEMENS SIRP program under data science/app development domain.
- Participated in Capgemini AI Agent Buildathon — national-level AI hackathon, approached customer support ai agent solution.
- Multilingual: English (Fluent), Tamil (Fluent), Hindi (Intermediate)